
NASA MOUSE BULK RNA-SEQ

Technical Report: Generative Models for NASA Mouse Bulk RNA-seq

DDIM, conditional WGAN-GP, GeneJEPA, harmonization, conditional generation, and downstream FLT/GC utility

EVIDENCE CUTOFF

28 July 2026

PRIMARY DATA

NASA OSDR API and mouse
ARCHS4

REPOSITORY ARTIFACT

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- **Status:** consolidated through 2026-07-28
- **Primary data:** NASA OSDR API-derived mouse bulk RNA-seq and mouse ARCHS4
- **Models evaluated:** Lacan et al. DDIM, Vinas et al. conditional WGAN-GP, and GeneJEPA

Executive Summary

The benchmark produced one defensible broad mouse generator and one defensible spaceflight-conditioned generator:

1. The paper-architecture Lacan DDIM trained on mouse ARCHS4 learned broad tissue-conditioned expression. Synthetic-trained tissue balanced accuracy was 0.869 on real held-out profiles, compared with 0.895 for real-trained classification. Its precision, recall, adversarial accuracy, and gene moments were strong, although gene-correlation agreement was 0.879 rather than the stricter 0.98 target.
2. The ARCHS4-pretrained, OSDR-adapted factorized DDIM passed every predeclared finite-sample fidelity gate on four locked-test generations. It can generate synthetic FLT or GC profiles for tissue, study, and material profiles represented during OSDR training. This is within-study interpolation, not generation for a novel accession.
3. The study-conditioned WGAN-GP did not pass joint fidelity. Its marginal support metrics were strong, but real and synthetic samples remained distinguishable with adversarial accuracy 0.636, outside the accepted 0.40 to 0.60 interval. Its locked test was therefore not opened.
4. GeneJEPA is a representation model with no expression decoder. The exact one-epoch mouse screen did not beat direct expression for tissue classification, so it was not advanced as a diffusion-guidance backbone. It cannot generate expression without adding a separate, non-paper GeneJEPA decoder.
5. Synthetic cohort augmentation did not improve FLT/GC prediction overall. On the locked OSDR test, real plus synthetic training reduced balanced accuracy from 0.754 to 0.734. A complete six-fold skeletal-muscle confirmation was also negative.
6. Generated data were more useful as a low-weight feature-selection view than as replacement biological samples. Independent held-out-study confirmation was strong in thymus and mixed in lung. The thymus result recovered a coherent FLT-down cell-cycle module with Reactome FDR below 0.05.

The practical conclusion is that DDIM is the best model in this benchmark. It is approved for controlled simulation within represented OSDR studies and for carefully validated feature guidance. It is not approved as a source of independent replicates, an unseen-study generator, or a default way to improve FLT/GC classifiers.

Scope And Questions

The work addressed three distinct questions that should not be conflated:

- **Broad expression generation:** can a model reproduce mouse bulk RNA-seq across tissues?
- **Conditional spaceflight generation:** can it generate realistic FLT and GC profiles for specified tissue and design covariates?
- **Downstream utility:** do generated profiles improve real-sample FLT/GC classification or help identify stable genes and pathways?

The benchmark included ARCHS4-only training, OSDR-only training, and ARCHS4 pretraining followed by OSDR adaptation. It tested pooled tissue-conditioned models, study conditioning, material conditioning, per-tissue behavior, multiple normalization and harmonization methods, direct synthetic augmentation, and generated-informed feature selection.

Evidence Policy

Results are interpreted in four evidence tiers:

Tier	Definition	Appropriate claim
Locked	Model and gates fixed before a one-time held-out evaluation	Model-level acceptance or rejection
Independent	Entire test accession removed from generator and classifier fitting	Evidence of transfer to that accession
Development	Repeated nested splits with accessions represented on both sides	Within-study screening and hypothesis generation
Historical	Earlier proxies, post-hoc tables, or models that failed a gate	Engineering provenance only

No scalar composite score is used. Correlation, precision, recall, F1, adversarial accuracy, Frechet distance, diversity, memorization, and conditional-effect recovery must pass independently. A high value on one metric cannot compensate for failure on another.

The earlier per-tissue "best use" table is a development artifact. It compared generated-only and real-plus-generated arms on validation data, including WGAN arms that subsequently failed model-level fidelity. It is not the final model-selection or biological-evidence table.

Data Provenance

NASA OSDR

OSDR expression and metadata were obtained through the repository's NASA OSDR API workflow. The deprecated integrated raw OSDR H5 file was not used.

- 1,631 API profile rows were found.
- 21 technical replicate rows were aggregated.
- The resulting matrix contains 1,610 biological profiles: 835 FLT and 775 ground control.
- Profiles span 75 OSDR accessions and 24 canonical material classes.
- The source matrix contains 48,694 shared mouse genes and RSEM expected counts.
- Tissue aliases such as `lvr` and `liver` were canonicalized before modeling.

The principal tissues were liver, skeletal muscle, skin, kidney, thymus, spleen, lung, and retina. Smaller API-returned tissues remained eligible for pooled conditioning and exploratory analyses. Skeletal-muscle material labels were retained where available.

The accepted conditional DDIM used a within-accession, tissue, and condition split: 781 training, 536 validation, and 293 locked-test profiles. Every represented test accession therefore had training examples. This design measures interpolation while controlling study context; it does not measure unseen-accession transfer.

ARCHS4

The local mouse ARCHS4 file contains 997,515 profiles and 53,511 genes. Metadata for the full file were inspected rather than treating the older 24,428-profile eight-tissue subset as the complete reference.

Three reusable cohorts were defined:

ARCHS4 cohort	Profiles	GEO series	Intended use
Control-only	23,614	3,213	Strict healthy/control sensitivity
Healthy-preferred	62,299	5,307	Primary reference pool
Broad	134,250	15,111	Diversity sensitivity

The paper-parity DDIM selected 17,244 unique healthy-preferred profiles across 20 tissue classes, split into 9,796 train, 2,448 validation, and 5,000 test profiles. GeneJEPA's bounded exact-architecture screen used 43,744 training exposures.

Model Implementations

Model	Paper-faithful core	NASA adaptation	Capability
Lacan ModelDDIM	Official 227M-parameter residual DDIM, 1,000 steps, paper optimizer and 15,000 epochs	Mouse landmarks; factorized OSDR condition adapters and calibration	Native expression generation
Vinas WGAN-GP	Released conditional generator/critic topology and training defaults	Mouse data; ARCHS4 pretraining and OSDR fine-tuning	Native expression generation
GeneJEPA	768-wide, 512-latent, 24-block predictive encoder	Mouse vocabulary and bulk profiles	Representation only

DDIM

The ARCHS4 model imports the official `ModelDDIM` implementation at inspected commit `cde890154698fcea96c924804aaff04af3351b48`. Its fixed configuration includes:

- two 8,192-unit residual layers and 227,109,786 parameters;
- 1,000 quadratic diffusion steps with beta 0.0001 to 0.02;
- summed noise-prediction MSE with antithetic timesteps;
- Adam at learning rate 0.0004783833151836702;
- OneCycle scheduling, AMP, EMA 0.999, and deterministic DDIM sampling;
- batch size 2,048 and 15,000 epochs.

Full-transcriptome TPM is computed before selecting a deterministic 974-gene mouse landmark panel. MaxAbs scaling is fitted on training profiles only.

The accepted OSDR extension adds factorized tissue, FLT/GC, study, and material conditioning through a rank-512 domain LoRA adapter. It uses a 4,000-step domain adaptation followed by a 1,000-step correlation-regularized refinement. Train-only calibration aligns global and shrunk accession/tissue means and adds partial positive missing-covariance residual noise. FLT/GC labels are not used to fit calibration means or covariances.

Conditional WGAN-GP

The WGAN uses the released Vinas et al. topology:

- 64-dimensional noise;
- two 256-unit ReLU layers in the generator and critic;
- learned embeddings for categorical covariates;
- RMSprop at `5e-4`;
- five critic updates per generator update;
- gradient penalty 10 and batch size 32.

The OSDR model conditions on tissue, FLT/GC, accession, material, sex, and muscle group where available. It is a conditional sampler. It is not a paired FLT-to-GC counterfactual model with subject-level identity preservation.

GeneJEPA

GeneJEPA encodes masked expression into a self-supervised representation. The public model exposes embeddings but no expression decoder. A GeneJEPA-guided generator would therefore require a separate diffusion or decoder and would be a new method, not paper-faithful GeneJEPA.

Evaluation Metrics

The main metrics answer different failure modes:

- **Correlation agreement:** similarity between real and synthetic gene-gene correlation structure. Higher is better.
- **Precision:** fraction of generated support that lies near real support. Low precision indicates unrealistic or over-broad samples.
- **Recall:** fraction of real support covered by generated support. Low recall indicates mode loss.
- **F1:** harmonic mean of precision and recall.
- **Adversarial accuracy (AA):** held-out real-versus-synthetic discrimination. Approximately 0.5 is desirable; the accepted interval was 0.40 to 0.60.
- **Frechet distance (FD):** distance between real and synthetic embedding distributions. Lower is better. Conditional evaluations normalize FD to a real-split reference.
- **Reverse validation:** train on synthetic, test on real.
- **FLT/GC effect recovery:** correlation and direction agreement between real and synthetic FLT-minus-GC gene effects.
- **Accession-aware recovery:** FLT-minus-GC effects estimated within each study and then combined, preventing between-study differences from being mislabeled as spaceflight biology.
- **LOO-FDR stability:** the FDR requirement must survive omission of each accession in turn. This is stricter than ordinary pooled FDR.

Generated rows are never counted as new biological replicates in FDR testing.

Broad Mouse Tissue Generation

The exact paper-architecture ARCHS4 DDIM completed 15,000 epochs and 75,000 optimizer steps on an A100 in 5,987 seconds, with peak allocated GPU memory of 5.93 GB.

Metric	ARCHS4 DDIM result	Interpretation
Real-train to real-test tissue BA	0.8954	Real-data ceiling for the probe
Synthetic-train to real-test tissue BA	0.8687	Strong tissue utility
Gene mean correlation	0.9965	Pass
Gene standard-deviation correlation	0.9437	Strong
Gene-correlation agreement	0.8791	Below strict 0.98 target
Direct precision	0.9655	Pass
Direct recall	0.8645	Pass finite-sample rule
Nearest-neighbor AA	0.512	Real and synthetic are difficult to separate
PCA-50 precision / recall	0.9855 / 0.9430	Strong embedded support
PCA-50 FD	0.0385	Low

The model learned tissue-specific generation despite overlap in the first two PCA components. The final two-dimensional PCA silhouette was -0.271, which is not inconsistent with the strong held-out tissue probe because a nonlinear or higher-dimensional boundary can retain information that PC1 and PC2 do not display.

The raw model-scale output contains 9.04% negative entries. Those values are valid in the unconstrained scaled model space but are not physical TPM. Any export for downstream expression analysis must use an explicit nonnegative clipping or nonnegative-generation policy and record it.

Decision: useful as a defensible broad 20-tissue mouse reference generator, but not accepted as an all-metric finalist because correlation-structure agreement missed the strict target. It is not by itself a spaceflight generator.

OSDR Conditional Generation

Accepted Factorized DDIM

Four independent synthetic cohorts were generated against the fixed 293-profile locked test.

Metric	Mean	Range	Repeats passing
Gene-correlation agreement	0.977	0.974 to 0.979	4/4
Precision	0.998	0.997 to 1.000	4/4
Recall	0.997	0.997 to 0.997	4/4
F1	0.997	0.997 to 0.998	4/4
Adversarial accuracy	0.458	0.454 to 0.464	4/4
FD / real-split P95	0.075	0.047 to 0.089	4/4

The test-size bootstrap correlation floor was 0.950, so all repeats pass the finite-sample rule. They remain just below the separate 0.98 paper target.

Pooled FLT/GC effect recovery passed three of four generations. Mean effect correlation was 0.598 and mean direction agreement was 0.683. The skeletal-muscle accession-aware diagnostic passed all four generations, with mean meta-effect correlation 0.608 and direction agreement 0.606 across five accessions. Exact jointly LOO-FDR-stable real/synthetic genes remained sparse: 0, 0, 0, and 1 across the four generations.

This combination matters. The model reproduces the broad conditional distribution and an aggregate FLT/GC contrast, but it does not reproduce a stable exact gene list.

Per-Tissue Validation Screen

The table below reports DDIM FLT/GC effect correlation, direction agreement, and repeat passes on shared validation data. It is useful for tissue prioritization but is below the locked evidence tier.

Tissue	Profiles / accessions	Tissue-level r / direction / pass	Accession-aware r / direction / pass
Liver	79 / 7	0.44 / 0.59 / 4/6	0.36 / 0.58 / 5/6
Skeletal muscle	61 / 11	0.57 / 0.77 / 6/6	0.24 / 0.58 / 3/6
Skin	50 / 6	0.72 / 0.71 / 6/6	0.62 / 0.69 / 6/6
Kidney	45 / 6	-0.41 / 0.34 / 0/6	-0.49 / 0.30 / 0/6
Thymus	37 / 4	0.68 / 0.72 / 6/6	0.76 / 0.63 / 6/6
Spleen	35 / 4	-0.22 / 0.45 / 0/6	-0.38 / 0.32 / 0/6
Lung	25 / 3	0.19 / 0.48 / 2/6	0.12 / 0.48 / 0/6
Retina	26 / 3	0.81 / 0.75 / 6/6	0.48 / 0.61 / 4/6

Skin and thymus were the strongest validation tissues after accession control. Retina was strong without accession adjustment but less stable after it. Liver was moderate. Kidney, spleen, and lung did not recover the contrast in this generation screen. The locked-test muscle result was stronger than its earlier validation screen, but exact gene overlap remained negligible.

Classifier Utility

Training data	Locked-test balanced accuracy	Locked-test ROC AUC
Real OSDR train	0.754	0.819
Synthetic train	0.700	0.751
Real plus synthetic	0.734	0.801

Synthetic profiles contain condition information, but adding them did not improve the broad real-only classifier.

Decision: accepted for conditional simulation inside represented studies. Rejected for unseen-study generation, exact gene replication, and default classification augmentation.

WGAN-GP Findings

The strongest study-conditioned WGAN validation result used the same 974-gene comparison space and six generation seeds.

Metric	Mean	Gate	Result
Correlation agreement	0.9759	repeat-specific floor 0.9756	4/6 pass
Precision	0.9764	at least 0.95	6/6 pass
Recall	0.9938	at least 0.85	6/6 pass
F1	0.9850	at least 0.90	6/6 pass
Adversarial accuracy	0.6362	0.40 to 0.60	0/6 pass
FD / real-split P95	0.1439	at most 1.0	6/6 pass

Twelve train-only calibration variants were evaluated. Added residual variance moved AA toward 0.5 but reduced correlation below its required floor. Per-gene quantile calibration also failed, indicating a joint-distribution mismatch rather than only a marginal-range problem.

Earlier WGAN transfer produced pooled FLT/GC effect correlation 0.805. That value was misleading in isolation: accession-aware correlation was -0.022 and zero of nine evaluable tissues passed. The WGAN could exploit pooled study-condition structure without reproducing the within-study FLT/GC effect.

Adversarial training does not guarantee that an external classifier reaches chance. The generator and training critic have finite capacity, optimize an approximate Wasserstein objective, and can stop before the full joint distribution is matched. External held-out discrimination is therefore still required.

Decision: rejected. The locked test remains unopened, and WGAN-generated profiles should not be used for augmentation or biological claims.

GeneJEPa Findings

The exact mouse screen used 4,096 train-selected highly variable genes and 43,744 ARCHS4 training exposures. It ran on an A100 with 31.68 GB peak allocated memory. One epoch required 1,074 training seconds and made only 238 optimizer updates, below the paper's 2,000-step EMA warmup.

Held-out tissue probe	Balanced accuracy	Macro F1
GeneJEPa embedding	0.703	0.701
Direct expression	0.839	0.840

Embedding and UMAP silhouettes were -0.176 and -0.215. The screen was much shorter than the paper's approximately 50 million profile exposures, so it is not evidence that a full-scale GeneJEPA model must fail. A paper-scale single-A100 run was estimated at roughly 20 GPU-days, however, and the bounded run did not justify that cost for this benchmark.

Decision: retain GeneJEPA only as an optional representation experiment. Do not describe it as a generator, and do not add a decoder under the GeneJEPA name without labeling the resulting method as new.

Harmonization Findings

A matched liver experiment held architecture, seed, 974 genes, training duration, and data split fixed while changing preprocessing. Each arm used 119 train and 50 validation profiles; the 70-profile OSD-379 test was not opened.

Method	Corr	Precision	Recall	F1	AA	FD ratio
No harmonization, TPM	0.283	0.200	0.960	0.331	0.850	2.052
lIangovan within-study z-score	0.278	0.160	1.000	0.276	0.770	0.716
Mentor two-stage z-score	0.348	0.440	1.000	0.611	0.690	0.977
ComBat	0.004	0.040	1.000	0.077	0.810	63.185
ComBat-seq	0.067	0.020	1.000	0.039	0.850	156.647
MBatch median polish	0.009	0.020	1.000	0.039	0.930	205.470
MBatch empirical Bayes	0.001	0.000	1.000	0.000	0.850	60.625
MBatch ANOVA	-0.003	0.020	1.000	0.039	0.870	44.716
MOBER	0.808	0.260	1.000	0.413	0.770	33.311

No method passed all fidelity criteria or either conditional-effect gate. MOBER's 0.808 correlation was the highest, but its low precision and F1, high AA, and 33-fold FD ratio show that the full distribution was not reproduced. The mentor two-stage transform was the closest balanced alternative, but it also failed.

This does not prove that harmonization is generally harmful. It shows that none of these adapters solved conditional generation on this fixed liver benchmark. ComBat and MBatch also remain transductive sensitivity methods for unseen batches, whereas MOBER is the principal inductive complex harmonizer tested here.

Downstream FLT/GC Utility

Direct Synthetic Augmentation

Three independent checks reached the same practical conclusion:

- Broad locked test: real plus synthetic BA 0.734 versus real-only 0.754.
- Adaptive per-tissue screen: heart improved on a five-profile, one-accession test; retina failed; skeletal muscle tied at a ceiling.
- Complete skeletal-muscle confirmation across 11 held-out accessions and 159 profiles: accession-macro BA changed from 0.655 to 0.658, AUROC from 0.718 to 0.690, and AP from 0.768 to 0.764. The BA bootstrap interval was -0.022 to 0.030.

Generated cohorts therefore do not provide a validated general augmentation gain. They must not be treated as added sample size in DGEA or pathway FDR.

Generated-Informed Feature Selection

The more successful use was to let one or more synthetic views influence feature ranking while keeping the final classifier anchored to real profiles or assigning synthetic rows very low total weight.

An independent confirmation removed OSD-900 lung and OSD-457 thymus from every OSDR generator fine-tuning role before evaluating them:

Tissue	Held-out study	Profiles	Baseline BA / AUC / AP	Guided BA / AUC / AP
Lung	OSD-900	20	0.400 / 0.450 / 0.523	0.550 / 0.550 / 0.635
Thymus	OSD-457	24	0.500 / 0.840 / 0.876	0.833 / 0.972 / 0.976

Across both studies, 11 predictions changed from incorrect to correct and none changed from correct to incorrect. Only one accession was tested per tissue, so this is not a study-level significance test.

The genotype audit strengthened thymus: metrics improved in both Nrf2KO and WT subgroups. Lung was mixed because KO AUROC fell from 0.56 to 0.52 even though its BA and AP improved.

The thymus signature contained **Birc5**, **Ccne2**, **Gmn**, **Ube2c**, **Cdk1**, **Nusap1**, **Ccnb1**, and **Ccnb2**, consistently FLT-down in both genotypes. Reactome terms for G2/M checkpoints, mitotic protein degradation, DNA replication, and cell-cycle control passed FDR 0.05. Lung highlighted **Cdkn1a**, **Ccne2**, **Pik3r3**, **Slc25a4**, **Mapk9**, and **Igfbp3**, but no lung Reactome term passed FDR 0.05.

Exploratory Within-Study Stability

Repeated nested development splits were run for 22 tissues. Because accessions were represented on both sides and the generator had seen the original training role, these are exploratory within-study results.

Tissue	Selected use	Mean BA / AUC / AP delta	Repeats all metrics nonworse	Real LOO-FDR stable genes
Kidney	Guided, low weight	+0.029 / +0.093 / +0.097	6/8	0
Liver	Guided, low weight	+0.024 / +0.054 / +0.044	5/8	0
Lung	Generated only	+0.086 / +0.156 / +0.157	7/8	0
Retina	Guided, low weight	+0.117 / +0.121 / +0.071	7/8	0
Skeletal muscle	Real plus generated	+0.043 / 0.000 / +0.013	5/8	7
Skin	Generated only	+0.086 / +0.109 / +0.065	4/8	0
Spleen	Guided, real only	+0.170 / +0.208 / +0.204	7/8	1
Thymus	Generated only	+0.121 / +0.092 / +0.075	7/8	8

The seven muscle LOO-stable genes were **Sox4**, **Sh3bp5**, **Cebpd**, **Cdkn1a**, **Bphl**, **Prkcd**, and **Arid5b**. The eight thymus genes were **Cenpe**, **Ccnb1**, **Nusap1**, **Stmn1**, **Cdk1**, **Top2a**, **Ccnb2**, and **Ccne2**. Spleen retained only **Igfbp3**.

The 292 significant Reactome rows produced across all tissues and overlapping gene sets are not 292 independent pathway discoveries. Reactome hierarchy, pathway overlap, and nested feature sets create extensive dependence.

What The Generator Can And Cannot Do

Supported

- Generate broad mouse tissue-conditioned expression from the ARCHS4 DDIM.

- Generate FLT or GC profiles for OSDR tissue, accession, and material combinations represented during factorized-DDIM adaptation.
- Reproduce the overall locked OSDR distribution with strong precision, recall, AA, and FD.
- Recover a moderate pooled FLT/GC effect and a locked skeletal-muscle accession-aware effect.
- Provide a useful secondary feature-ranking view, with the strongest independent evidence currently in thymus.

Not Supported

- Generation for a new OSDR study with no training representation.
- Treating generated profiles as independent animals or biological replicates.
- Exact recovery of a stable FLT/GC gene list.
- General improvement from adding large synthetic cohorts to classifier training.
- Subject-paired FLT-to-GC counterfactuals with all non-condition attributes fixed.
- WGAN use for final generation or biological interpretation.
- GeneJEPa expression generation without a new decoder.

Recommended Operating Protocol

1. Freeze the accepted factorized DDIM and its preprocessing for reported within-study simulation. Do not choose new calibration or hyperparameters using the locked test.
2. For downstream FLT/GC analysis, use generated profiles for feature ranking or weak regularization only. Fit the final model primarily on real profiles.
3. Evaluate every biological claim with within-accession real-data effects, random-effects meta-analysis, ordinary FDR, and leave-one-accession-out sensitivity. Synthetic rows must not enter the biological replicate count.
4. Keep OSD-900 and OSD-457 untouched for further tuning. Replicate the thymus feature policy in a newly obtained thymus accession and treat lung as exploratory until its direction is stable across genotype and study.
5. Extend API metadata ingestion to capture genotype, sex, material, and muscle group directly. Avoid reconstructing these fields from profile names where the source API or repository metadata can provide them.
6. Declare the nonnegative export policy for DDIM outputs. Store both the native scaled output and the clipped inverse-scaled expression with provenance.
7. For true unseen-study generation, design a new accession-held-out adaptation benchmark. Study conditioning should include an explicit unknown-study or hierarchical study representation rather than reusing a known accession label.
8. Do not spend full-scale GeneJEPa compute unless a representation task first demonstrates value over direct expression on a fresh benchmark.

Reproducibility And Artifact Index

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- DDIM versus WGAN tissue comparison
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The older reduced diffusion proxy, scalar-composite rankings, pooled WGAN effect correlation, and per-tissue "best use" table remain useful for provenance but are superseded for final decisions by the paper-parity DDIM, independent metric gates, locked conditional test, complete augmentation confirmation, and independent feature-guidance confirmation summarized here.